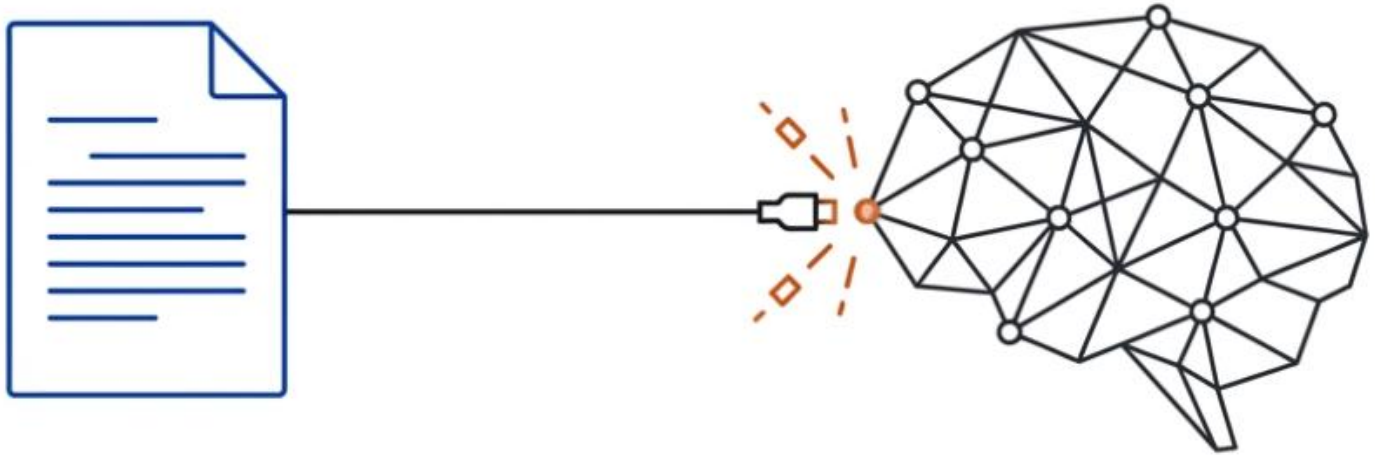


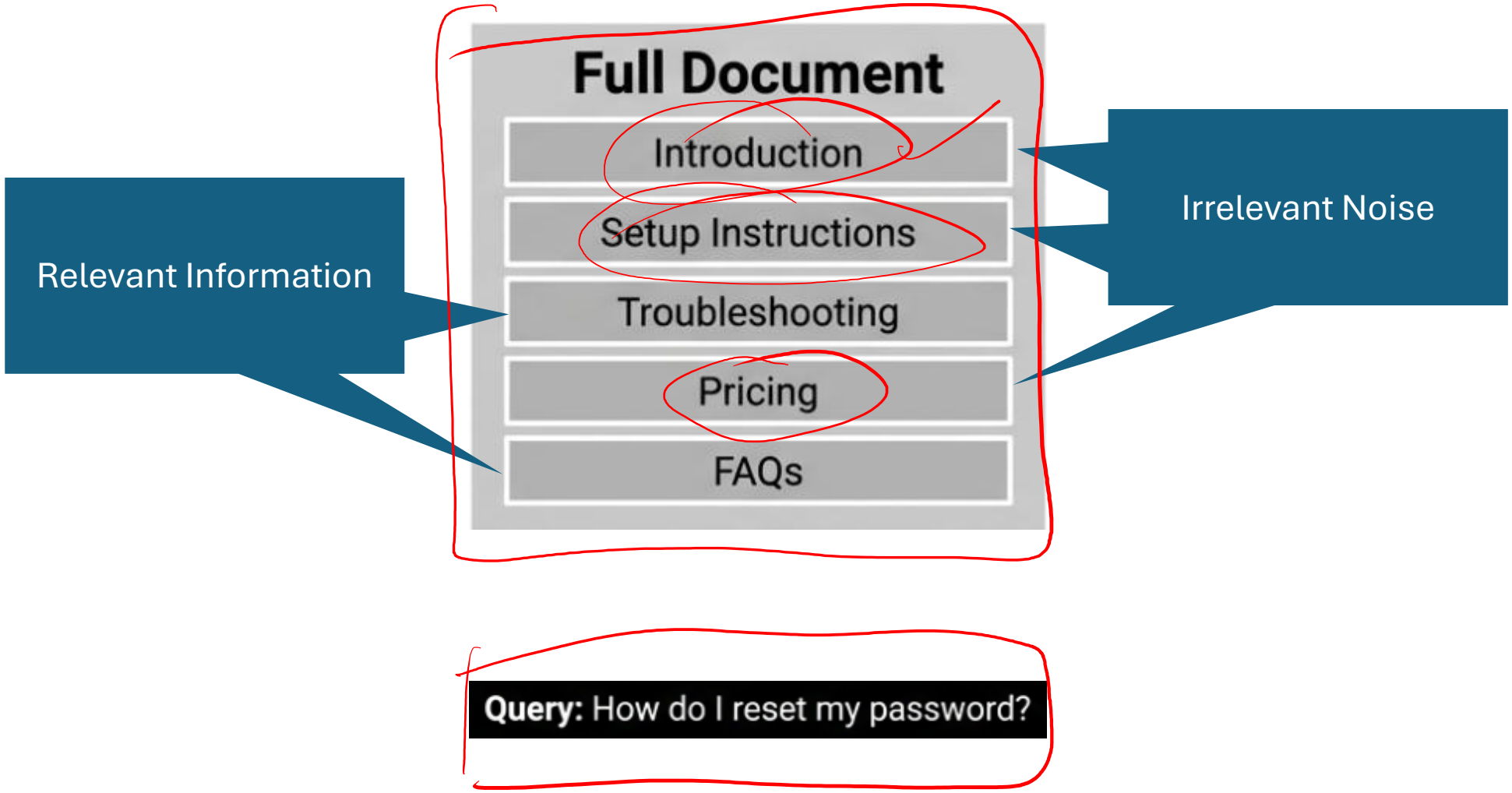
Making Retrieval Meaningful: From Raw Text to Useful Context

Chunking Strategies: fixed, semantic + Structure-aware + metadata design. Optimize chunk size/overlap, ChromaDB

https://tinyurl.com/rag-augmented

The picture till now...

The Core Problem: Context vs. Noise

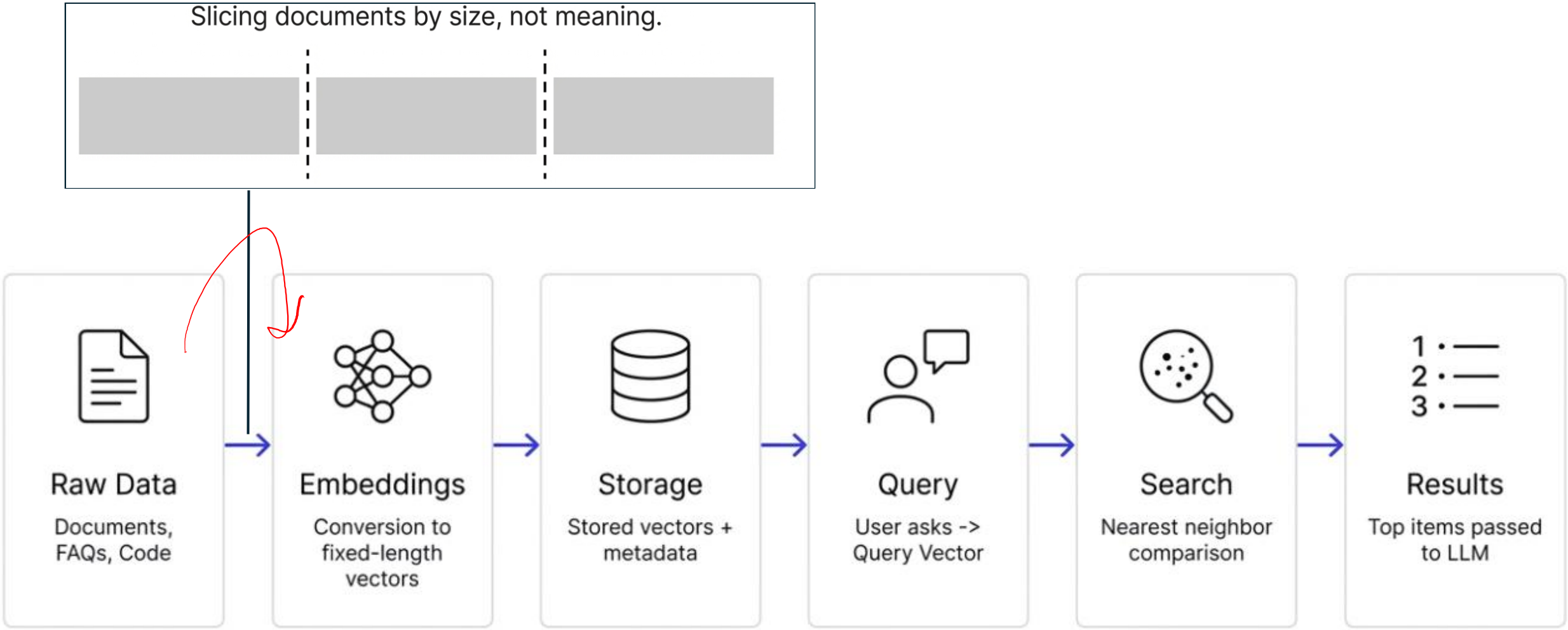


Key Takeaway: Chunking is the process of deciding the fundamental unit of retrieval. Retrieving entire documents overwhelms the model with irrelevant context.

Chunking helps not only accuracy, but also efficiency:

- better retrieval precision
- less irrelevant context
- lower token usage
- lower API cost
- faster responses

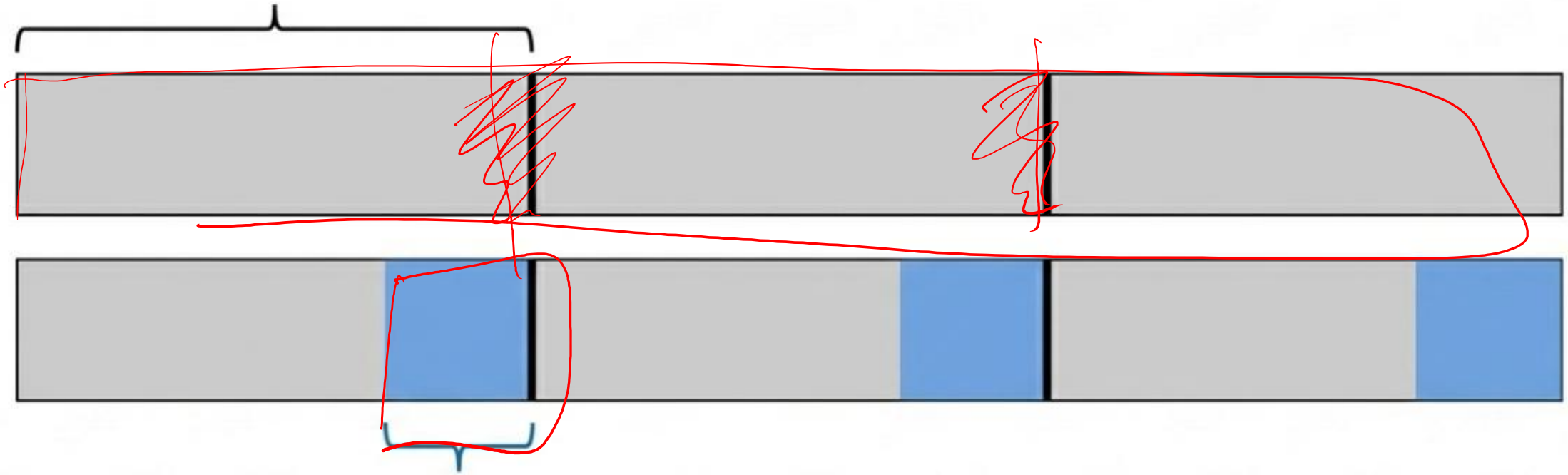
Chunking


Slicing documents by size, not meaning.
Raw Data
Documents, FAQs, Code
Embeddings
Conversion to fixed-length vectors
Storage
Stored vectors + metadata
Query
User asks -> Query Vector
Search
Nearest neighbor comparison
1
2
3
Results
Top items passed to LLM

The Anatomy of a Chunk: Size & Overlap

250

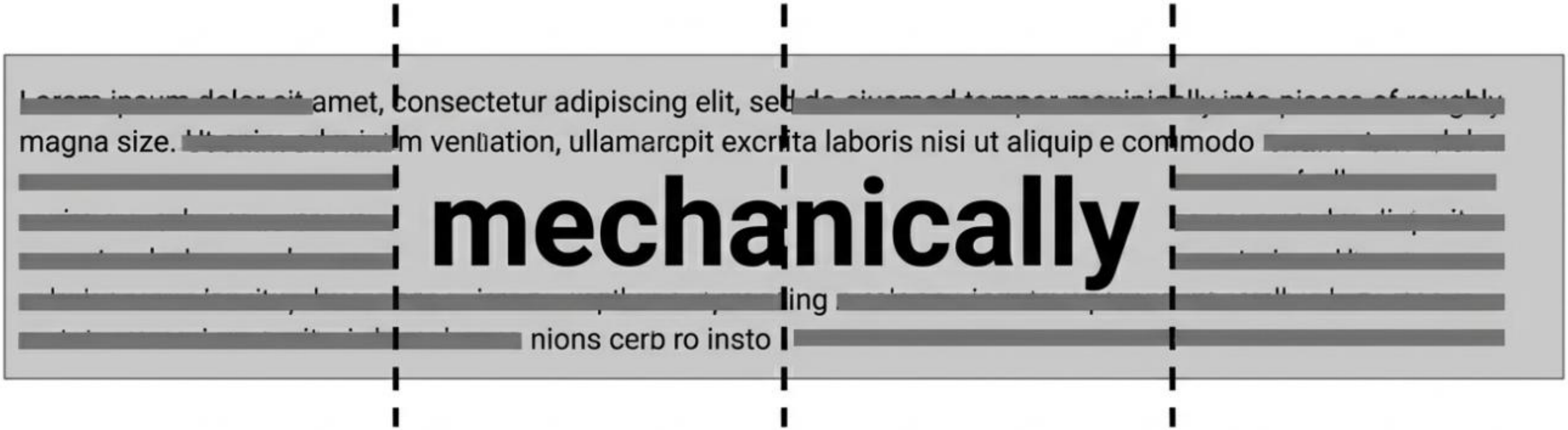
Chunk Size (e.g., 300 tokens)
The strict maximum capacity limit.



Overlap (e.g., 50 tokens)
The **safety net**: tokens repeated to stitch context across cuts.

What is Fixed Chunking?

512 , 512 , 512



Fixed chunking is the simplest and most common strategy. Documents are split mechanically into pieces of roughly equal size.

The Golden Rule: It does not ask if a topic changed, if a heading started, or if a paragraph ended. It only asks: Have we reached the size limit?

Fixed chunking does not care about meaning. It only cares about length.

Fixed Chunking: Choosing the Unit of Measurement

512

Characters	Words	Tokens
Split every N characters.	Split every N words.	**The Standard**
Crude. Cuts words awkwardly mid-letter. Does not map to LLM behavior.	Better readability, but embedding models and language models do not measure constraints in words.	Aligns perfectly with language model input limits and pipeline mechanics. The universally preferred baseline.

Highly precise, focused retrieval.

Trade-off: Severe context loss; risk of splitting simple explanations.

Balances complete context for multi-sentence answers without overwhelming the model.

Preserves broader narrative context.

Trade-off: Low precision; passes massive amounts of irrelevant noise to the model.

Core Principle: Smaller chunks isolate facts; larger chunks preserve narratives. The optimal size depends entirely on the expected length of the user's answer.

Fixed chunking splits documents by length, destroying context

Fixed chunking is a strong baseline, but it has one major limitation: it counts tokens, completely ignoring the internal meaning of the document.

To create a new user account, navigate to the portal and enter your secure email address. Example: If your company email is john@corp.com, use that to register. What happens if you forget your password? Click the reset link on the main page to receive a recovery token via email.

Cuts in the middle of a sentence

Divides a definition from its example

Separates a question from its answer

Semantic chunking preserves coherent units of thought

Instead of enforcing an arbitrary token limit, semantic chunking identifies where a topic meaningfully changes and creates boundaries there.

Instead of saying: Split every 300 tokens...

We say: Keep sentences together while they discuss the same idea, and **start a new chunk** when the topic shifts.

> “A chunk should ideally contain text that belongs together conceptually.”

Intro → Chunk 1

FAQs section: Multiple topics

Account Setup C2

File Upload Limits C3

Password Reset .....

Sharing Files .....

Billing .....

Highly vulnerable to topic-mixing if split by fixed token counts.

Semantic chunking outputs conceptually clean targets

By finding the natural boundaries in the text, each chunk becomes a **focused, meaningful** retrieval unit.

Semantic Chunk 1 [Account Setup]

...create a password, and verify your email... After verification, log in to access your dashboard.

Semantic Chunk 2 [File Upload Limits]

Free users can upload files up to 100 MB each. Premium users can upload files up to 5 GB each...

Semantic Chunk 3 [Password Reset]

If you forget your password, click the Forgot Password link...

Query Comparison: “What should I do if I forget my password?”

Baseline Fixed Chunk

...Premium users can upload files up to 5 GB each. If an upload fails, check your file size... If you forget your password, click the Forgot Password link on the login page...

Contains the answer, but drags in irrelevant upload noise.

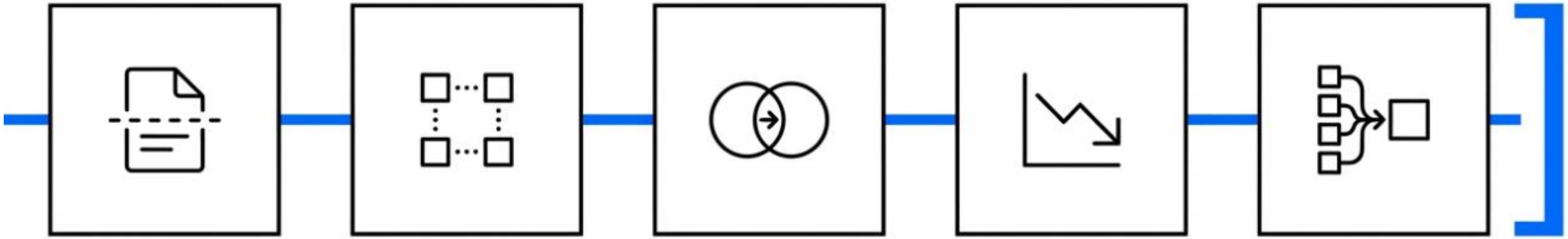
Semantic Chunk

If you forget your password, click the Forgot Password link on the login page. Enter your registered email address. A password reset email will be sent to you.

Clean, focused, self-contained evidence.

The mechanics: A five-step boundary detection process

Semantic chunking relies on computing the underlying meaning of small text units and measuring the distance between them.



1. Split

Break document into base units (sentences/short blocks).

2. Embed

Compute vector embeddings for each unit to represent meaning.

3. Compare

Check similarity scores between neighboring units.

4. Detect

Identify significant drops in similarity (the topic shift).

5. Group

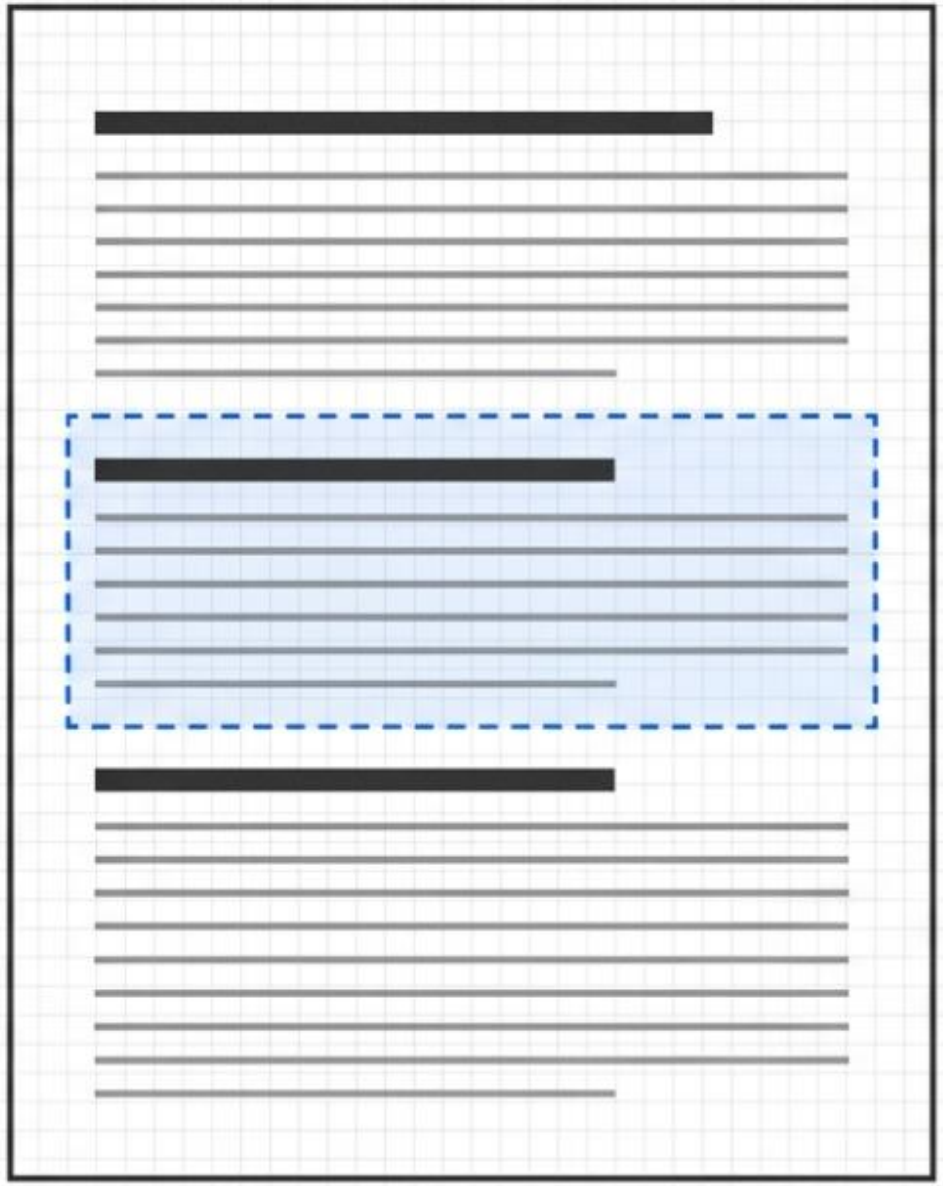
Merge similar adjacent units into a single chunk.

Structured Chunking

The document already tells us how to split it.

- Rather than forcing arbitrary boundaries based on token counts, we can respect the way the document was originally organized.
- Authors use titles, headings, bullet lists, and paragraphs to signal distinct topics.
- A structure-aware chunker uses these explicit boundaries as the guide for chunk creation.

The Authors Intent Wireframe



Preserving the Question-Answer Pair

Fixed chunkers risk separating the question from its answer, destroying context. Structure-aware chunkers naturally keep FAQ items intact. This is highly retrieval-friendly because user queries often perfectly match the FAQ question itself.

Fixed

Q: How do I reset my password?

A: Click the Forgot Password link on the login page. Enter your email address, and follow the link sent to your inbox.

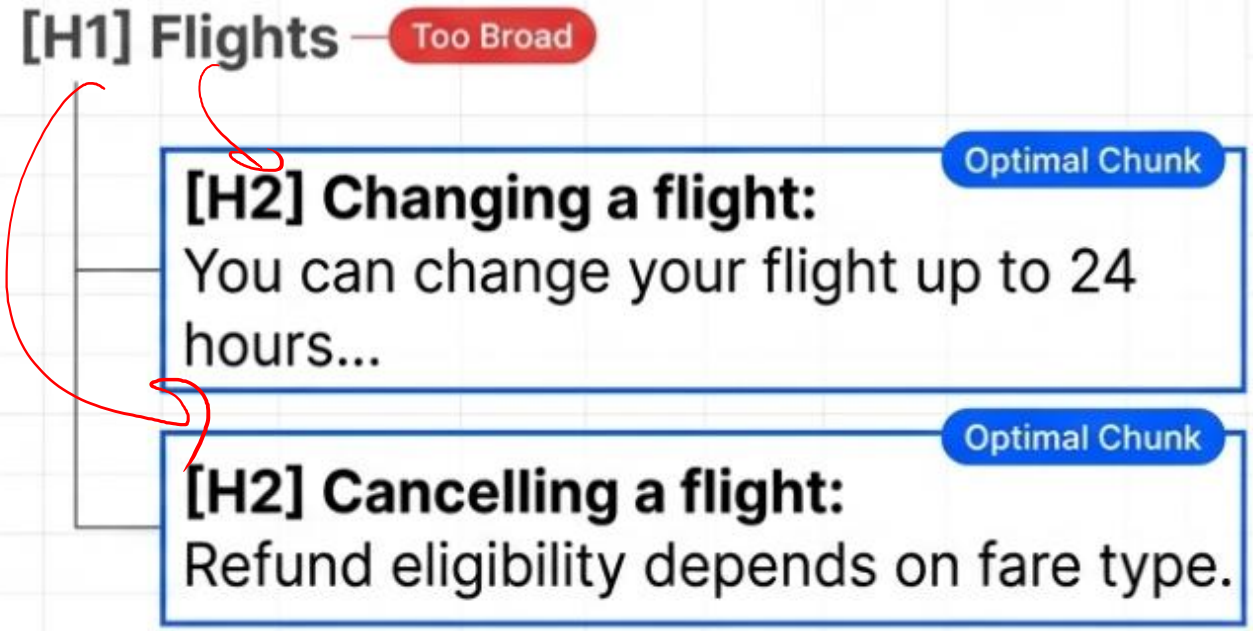
Structure-Aware

Q: How do I reset my password?

A: Click the Forgot Password link on the login page. Enter your email address, and follow the link sent to your inbox.

Hierarchy matters: Choose the right level of structure.

- Chunking at the top-level heading mixes distinct user intents.
- Chunking at the subheading level isolates distinct actions.
- Structure-aware chunking requires identifying the optimal structural depth.



A chunk is not just text.

The Naked Chunk (Text Only)

If you forget your password, click the Forgot Password link on the login page. Enter your registered email address. A password reset email will be sent to you.

The Enriched Chunk (Text + Metadata)

source: cloud_storage_guide.pdf

section: Password Reset

If you forget your password, click the Forgot Password link on the login page. Enter your registered email address. A password reset email will be sent to you.

chunk_index: 7

doc_type: user_guide

language: en

The text tells us **what the chunk says**. The metadata tells us **what the chunk is and where it came from**.

Why Metadata is Not Optional

1. Retrieval Control

Restrict searches to specific subsets (e.g., only search `doc_type: faq`).

2. Interpretability

Understand exactly why a specific chunk was retrieved by the system.

3. Grounding & Citations

Enable the LLM to cite verifiable sources (e.g., 'According to page 3...').

4. Debugging

Turn vague retrieval errors into analyzable system behavior patterns.

5. Routing & Filtering

Create boundaries based on user access or strict business domains.

The Taxonomy of Metadata

Source

Where it came from. Useful for citations and traceability.

`source: hr_policy.pdf`

Structural

Where it sits inside the document. Aligns structure with meaning.

`section: Leave Requests`

Content-type

What kind of content it is. Helps filter or rank differently.

`doc_type: faq`

Domain

What business area it belongs to. Crucial for enterprise routing.

`product: cloud_storage`

Temporal

When it was created or valid. Ensures users get the latest guidance.

`updated_at: 2026-03-10`

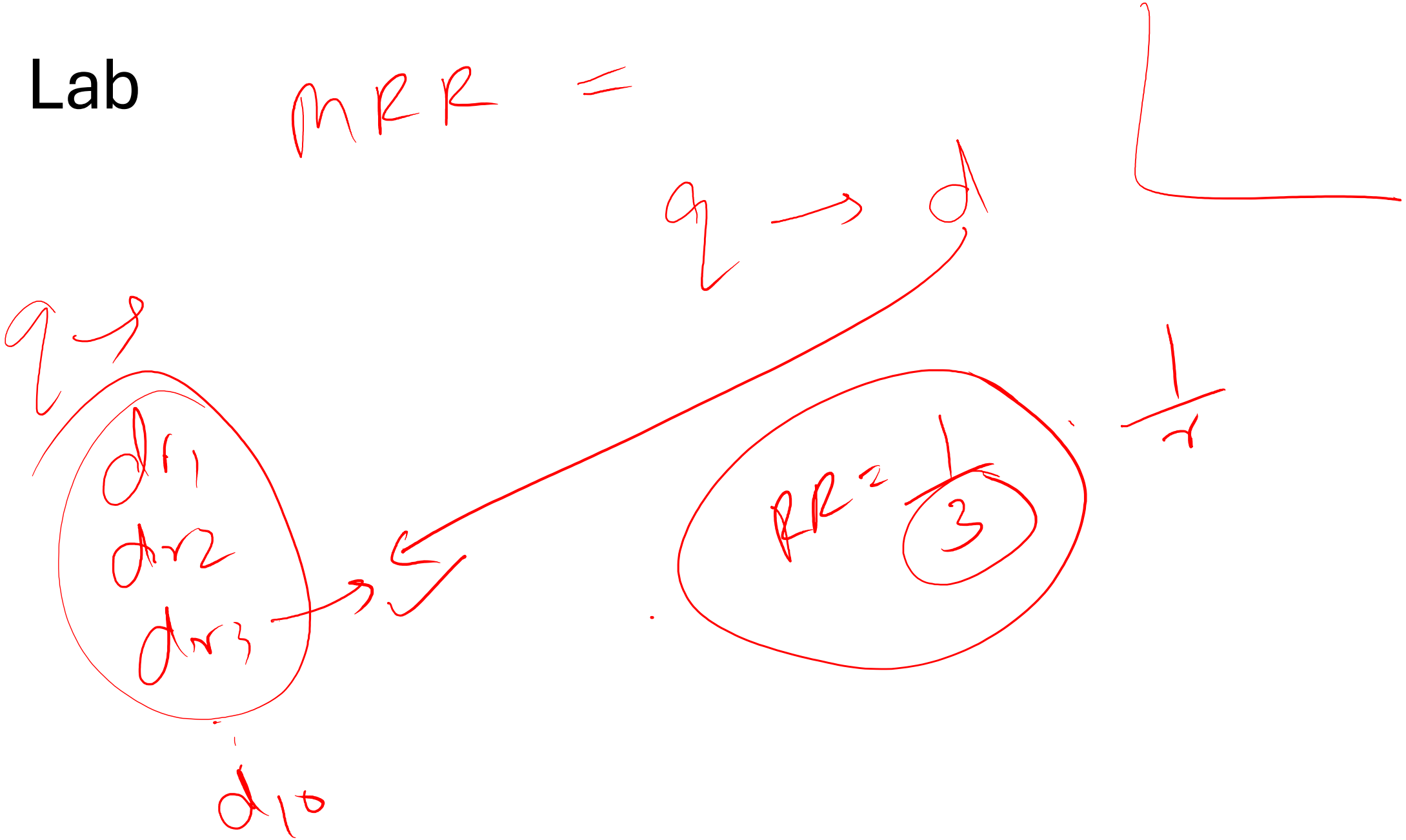
Access

Who is allowed to see it. Enforces retrieval boundaries.

`access_level: internal`

Lab

MRR =



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Thank You

Appendix